| | | | |
|---|---|---|---|
| Tomatoes (can) | 300 | 237,000 | 790 |

Then you try to go top to bottom, picking up the food with the highest ratio of calories per unit of weight, ending up with peanut butter, rice, wheat flour and strawberry jam, for a total of **2,813,400 calories**.

That's much better than our first result. It's clear that this was a step in the right direction. But, looking closely, it's apparent that adding peanut butter prevented us from taking beans, which would have allowed us to increase the total value of the crate even more. The good news, is that at least you won't be forced to the Martian's diet anymore: no potatoes goes on Mars this time.

After a few more hours of brainstorming, you are all pretty much ready to give up: the only way you can solve this is to check, for each item, if you can get a better solution by including it or leaving it out. And the only way you know for doing so, is enumerating all possible solutions, filter out the ones that are above the weight threshold, and pick the best of the remaining ones. This is what's called a *brute force* algorithm and we know from Math that it's a very expensive one.

Since for each item you can either pack it or leave it, the possible solutions are $2^7=128$. You guys are not too keen on going through a hundred solutions, but after a few hours, despite understanding why it's called brute force, you are totally exhausted, but almost done with it.

Then the news breaks: someone called from the mission control, and following up on complaints from some settlers-